

TASK: Library Database (SQL)

Create Database

```
CREATE DATABASE LibraryDB;  
USE LibraryDB;
```

Create BOOKS Table

```
CREATE TABLE Books(  
    book_id INT PRIMARY KEY,  
    title VARCHAR(50)  
);
```

Create MEMBERS Table

```
CREATE TABLE Members(  
    member_id INT PRIMARY KEY,  
    name VARCHAR(50)  
);
```

Create ISSUE RECORDS Table (with Foreign Keys)

```
CREATE TABLE IssueRecords(  
    issue_id INT PRIMARY KEY,  
    book_id INT,  
    member_id INT,  
    FOREIGN KEY(book_id) REFERENCES Books(book_id),  
    FOREIGN KEY(member_id) REFERENCES Members(member_id)  
);
```

Check Tables Created

SHOW TABLES;

Insert Sample Data (For Demo)

Insert Books

INSERT INTO Books VALUES

(1,'Java'),
(2,'Python'),
(3,'Data Science');

Insert Members

INSERT INTO Members VALUES

(101,'Arun'),
(102,'Meena'),
(103,'Karthik');

Issue a Book

Now link Book → Member

INSERT INTO IssueRecords VALUES

(1,1,101),
(2,2,102);

Meaning:

Issue ID	Book ID	Member ID
1	1	101

View Issued Books

```
SELECT
IssueRecords.issue_id,
Books.title,
Members.name
FROM IssueRecords
JOIN Books ON IssueRecords.book_id = Books.book_id
JOIN Members ON IssueRecords.member_id = Members.member_id;
```

Issue ID	Book	Issued To
1	Java	Arun
2	Pytho n	Meena

Modify IssueRecords (Add Dates for Overdue Logic)

Drop old table first:

```
DROP TABLE IssueRecords;
```

Create with dates:

```
CREATE TABLE IssueRecords(
  issue_id INT PRIMARY KEY,
  book_id INT,
  member_id INT,
  issue_date DATE,
  return_date DATE,
  FOREIGN KEY(book_id) REFERENCES Books(book_id),
  FOREIGN KEY(member_id) REFERENCES Members(member_id)
);
```

Insert 10 Records

Insert Books

```
INSERT INTO Books VALUES
(11,'Java Programming'),
(12,'Python Basics'),
(13,'Data Structures'),
(14,'Machine Learning'),
(15,'Database Systems'),
(16,'Operating Systems'),
(17,'Cloud Computing'),
(18,'AI Fundamentals'),
(19,'Networking'),
(20,'Cyber Security');
```

Insert Members

```
INSERT INTO Members VALUES
(201,'Arun'),
(202,'Meena'),
(203,'Karthik'),
(204,'Divya'),
(205,'Rahul'),
(206,'Sneha'),
(207,'Vijay'),
(208,'Anita'),
(209,'Ravi'),
(210,'Priya');
```

Insert Issue Records

```
INSERT INTO IssueRecords VALUES
(1,11,201,'2026-02-01',NULL),
(2,12,202,'2026-02-05','2026-02-15'),
(3,13,203,'2026-01-20',NULL),
```

(4,14,204,'2026-02-10',NULL),
(5,15,205,'2026-01-25','2026-02-01'),
(6,16,206,'2026-02-12',NULL),
(7,17,207,'2026-01-15',NULL),
(8,18,208,'2026-02-18',NULL),
(9,19,209,'2026-02-20',NULL),
(10,20,210,'2026-01-30','2026-02-10');

Required Queries

1. Display All Available Books

```
SELECT b.book_id, b.title
FROM Books b
LEFT JOIN IssueRecords i ON b.book_id = i.book_id
WHERE i.return_date IS NOT NULL OR i.issue_id IS NULL;
```

2. Find Books Issued by Specific Member

Example: Arun (101)

```
SELECT b.title
FROM IssueRecords i
JOIN Books b ON i.book_id = b.book_id
WHERE i.member_id = 101;
```

3. Count Total Books Issued

```
SELECT COUNT(*) AS Total_Issued
FROM IssueRecords;
```

4. Show Overdue Books

```
SELECT b.title, m.name, i.issue_date
FROM IssueRecords i
```

```

JOIN Books b ON i.book_id = b.book_id
JOIN Members m ON i.member_id = m.member_id
WHERE i.return_date IS NULL
AND DATEDIFF(CURDATE(), i.issue_date) > 7;

```

Create View for Currently Issued Books

```

CREATE VIEW Current_Issued_Books AS
SELECT b.title, m.name, i.issue_date
FROM IssueRecords i
JOIN Books b ON i.book_id = b.book_id
JOIN Members m ON i.member_id = m.member_id
WHERE i.return_date IS NULL;

```

Check view:

```

SELECT * FROM Current_Issued_Books;

```

```

+----+-----+-----+
| 2 | Python |
| 3 | Data Science |
| 12 | Python Basics |
| 15 | Database Systems |
| 20 | Cyber Security |
+----+-----+-----+
6 rows in set (0.00 sec)

mysql> SELECT b.title
-> FROM IssueRecords i
-> JOIN Books b ON i.book_id = b.book_id
-> WHERE i.member_id = 101;
Empty set (0.01 sec)

mysql> SELECT COUNT(*) AS Total_Issued
-> FROM IssueRecords;
+-----+
| Total_Issued |
+-----+
| 10 |
+-----+
1 row in set (0.01 sec)

mysql> SELECT b.title, m.name, i.issue_date
-> FROM IssueRecords i
-> JOIN Books b ON i.book_id = b.book_id
-> JOIN Members m ON i.member_id = m.member_id
-> WHERE i.return_date IS NULL
-> AND DATEDIFF(CURDATE(), i.issue_date) > 7;
+-----+-----+-----+
| title | name | issue_date |
+-----+-----+-----+
| Java Programming | Arun | 2026-02-01 |
| Data Structures | Karthik | 2026-01-20 |
| Machine Learning | Divya | 2026-02-10 |
| Operating Systems | Sneha | 2026-02-12 |
| Cloud Computing | Vijay | 2026-01-15 |
+-----+-----+-----+
5 rows in set (0.01 sec)

mysql> CREATE VIEW Current_Issued_Books AS
-> SELECT b.title, m.name, i.issue_date
-> FROM IssueRecords i
-> JOIN Books b ON i.book_id = b.book_id
-> JOIN Members m ON i.member_id = m.member_id
-> WHERE i.return_date IS NULL;
Query OK, 0 rows affected (0.01 sec)

mysql> SELECT * FROM Current_Issued_Books;
+-----+-----+-----+
| title | name | issue_date |
+-----+-----+-----+
| Java Programming | Arun | 2026-02-01 |
| Data Structures | Karthik | 2026-01-20 |
| Machine Learning | Divya | 2026-02-10 |
| Operating Systems | Sneha | 2026-02-12 |
| Cloud Computing | Vijay | 2026-01-15 |
| AI Fundamentals | Anita | 2026-02-18 |
| Networking | Ravi | 2026-02-20 |
+-----+-----+-----+
7 rows in set (0.00 sec)

mysql>

```